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Roll No. .....

337614(37)

B. E. (Sixth Semester)
EXAMINATION, April-May, 2015
(Old Scheme)

(Branch : Mechanical)

MACHINE TOOL TECHNOLOGY

Time : Three Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks : 28*

Note : Answer all questions. Give neat and clean figures at respective places, if needed. Part (a) of each question is compulsory, while answer any *two* out of the Parts (b), (c) and (d). Assume suitable data if missing.

1. (a) What precautions are to be required to avoid built-up-edge formation ? 2

   (b) How does a single point cutting tool is designated in ASA system and ORS system ? Give complete descriptions of various angles in both the system giving their diagrammatic representation. 7

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   (c) Discuss the assumptions and salient features of Earnst-Merchant theory and Lee and Shaffer theory.

   (d) The following data were obtained during an orthogonal turning operation on a lathe : cutting force 120 kg, feed force 30 kg, back rake angle 15°, feed 0.2 mm per revolution, cutting speed 100 m per minute, depth of cut 0.4 mm, chip thickness 0.3 mm. The diameter of workpiece is 120 mm. Calculate : chip thickness ratio, shear angle, coefficient of friction, angle of friction, shear stress, shear strain, strain energy and chip flow velocity. 7

2. (a) Differentiate between machinability and machinability index. 2

   (b) Discuss the heat generation during metal cutting in the following regions : around shear plane, around tool-chip interface and around tool work interface. Also discuss salient effects of heat generated on the tool and the workpiece. 7

   (c) During a turning operation of a workpiece on a lathe, following data were recorded : cutting speed 25 m per minute, feed 0.3 mm per revolution, depth of cut 2 mm, tool life 100 minutes. For this operation, the tool life equation is given as : 7

$$VT^{0.12} f^{0.7} d^{0.3} = C$$

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If the cutting speed, feed and depth of cut are all increased by 25% each and also collectively, what will be their effect on tool life ?

(d) How do you differentiate coolant, lubricant and cutting fluid from each other ? Further, for the following operations, give the primary cutting conditions and requirements and suggest suitable fluids. The operations are : turning, drilling, reaming, broaching and milling. 7

3. (a) What is the main reason of bending in lathe bed and how it can be avoided or optimized ? 2

(b) Discuss the effects of the following in the design of machine tool structures : stiffness and rigidity, material of structures, types of cross-section of the structures and wear-resistance of structures. Give suitable diagrams wherever needed. 7

(c) What is torsional rigidity of lathe bed ? Why is it important in designing the structure of lathe ? How it may affect the performance and accuracy of the lathe operations ? 7

(d) What are the factors and parameters considered for designing antifriction guideways ? Also list some of the main advantages and drawbacks of anti-friction guideways. 7

4. (a) What is "speed loss" ? How can it be avoided ? 2

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(b) What is "machine tool drive" ? What are different types of drives employed for mach tool operations ? Give suitable diagrams when necessary. Also discuss their merits in brief.

(c) Explain all about ray-diagram, saw-diagram speed-diagram giving their respective figu What are the uses or advantages of the diagrams ?

(d) Draw only speed-ray diagram for a head-stock speed gear box for speeds ranging from 45 r. p. to 2000 r. p. m. with $\phi = 1.25$. Show all the calculations and assumptions considered, if any.

5. (a) List the names of the operations involved in the installation of machine tools.

(b) Explain in detail the Norton drive, its structure and working giving suitable neat and clean figure. Also list its advantages and drawbacks.

(c) Explain preventive maintenance, corrective maintenance and re-conditioning of machine tools. How do they differ from each other ? Give suitable examples of each of them.

(d) Design a feed gear box for metric thread cutting operation which consists 15 different pitches ranging from 1 mm to 6 mm. Assume suitable data if needed and clearly mention it in your answer with justification.

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